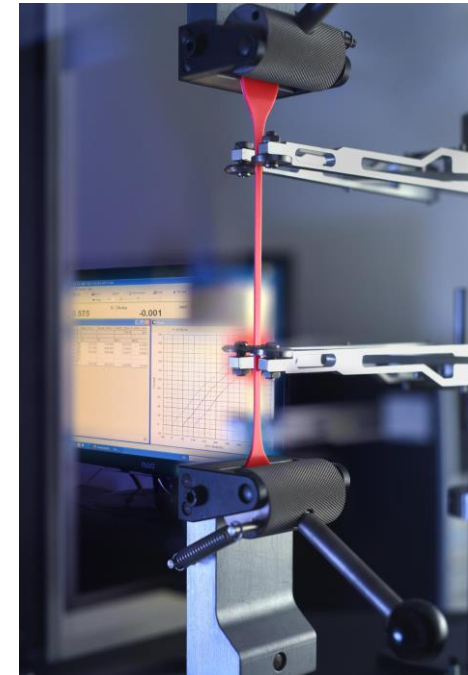


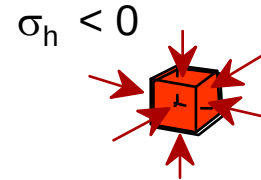
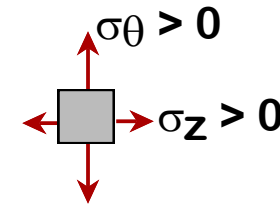
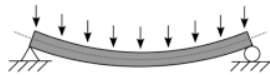
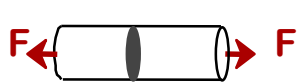
TOPIC 5. MECHANICAL PROPERTIES I

- **Basic Concepts**
- **Uniaxial Tension Test**
- **Stress-strain Curve: Elastic Zone**
- **Stress-strain Curve: Plastic Zone**
- **Slip Systems**
- **Stress-strain Curve: Calculations**
- **True Stress-strain Curves**

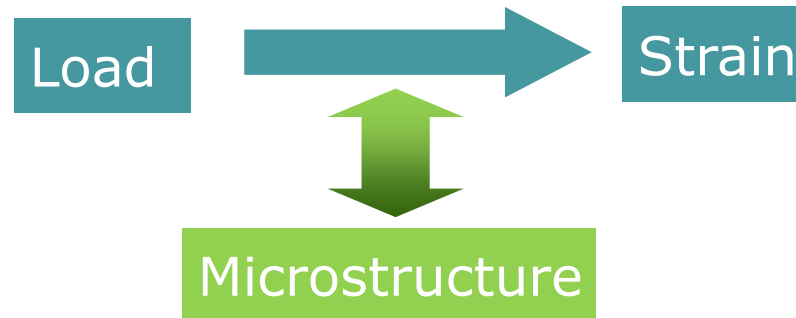


BASIC CONCEPTS

Mechanical property: how a material responds to an outside stimuli induced by mechanical forces



Mechanical properties determine the behaviour of materials $\Rightarrow$ determine applications.

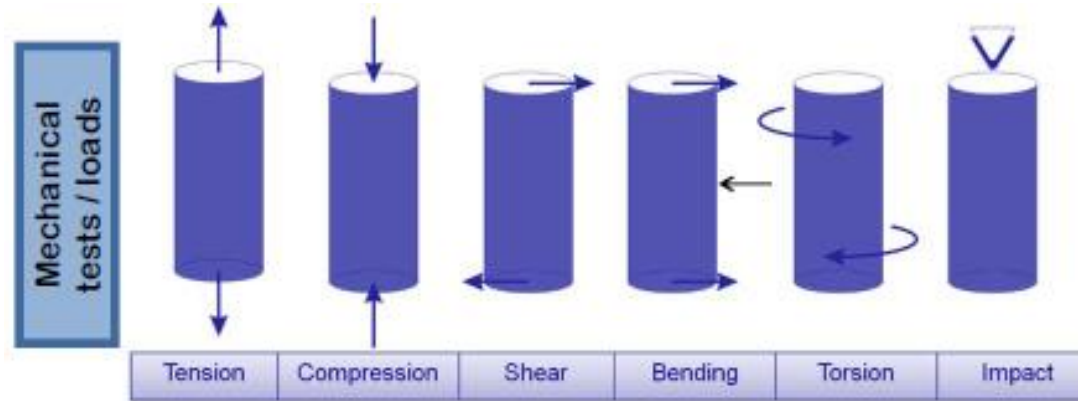


Key Mechanical properties: stiffness, strength, hardness, ductility and toughness

BASIC CONCEPTS

How is the response of a material to a F_{applied} measured?

TESTS



Performing carefully designed (standardized) laboratory experiments that replicate as nearly as possible the service conditions (**applications**).

Tests are **standardized**: obey certain established **standards**

→ TESTS

Service Conditions

- Determine materials properties and determine their possible uses.
- Quality control in the fabrication process.
- Determine the previous history and composition of materials.
- Establish the failure causes of the material during service
- Study new type of materials

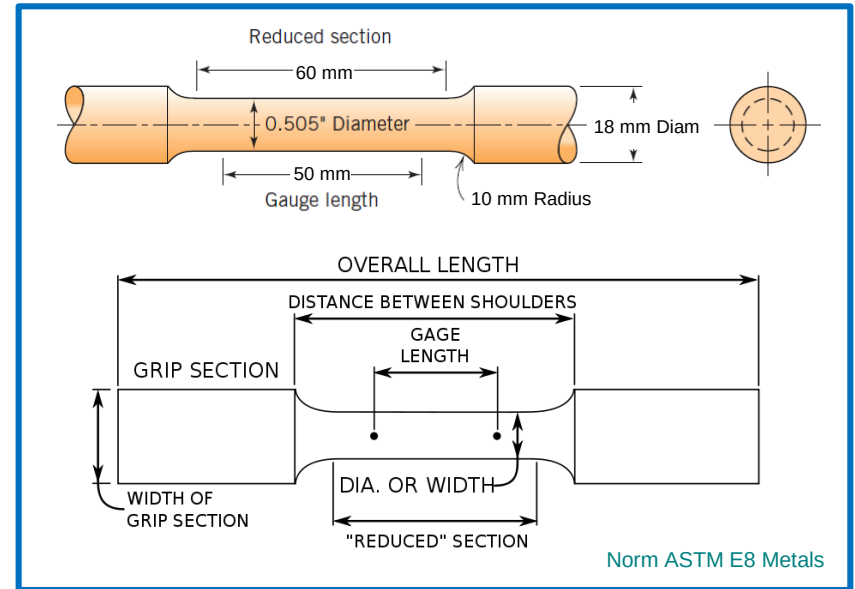
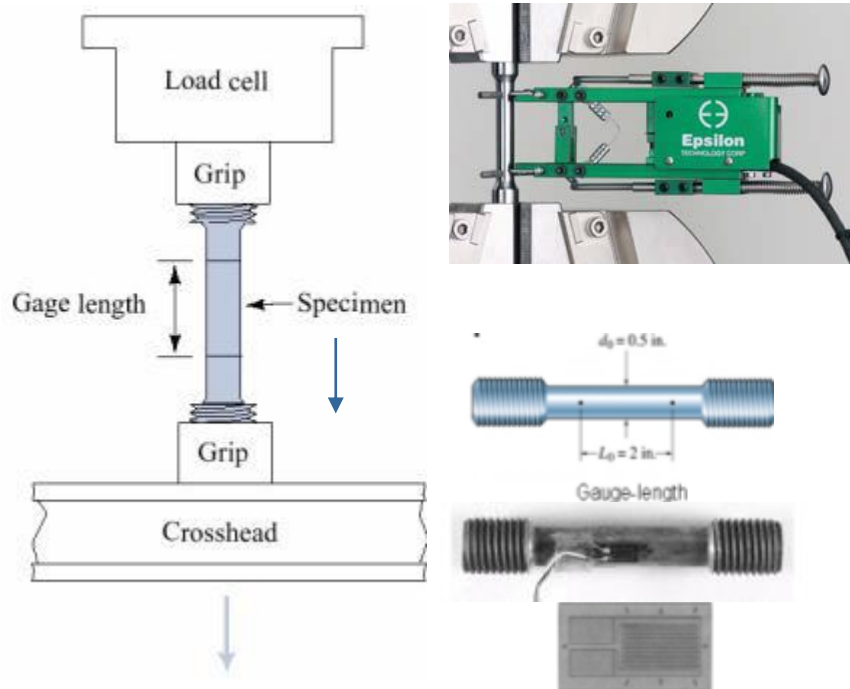
BASIC CONCEPTS

Uniaxial tension test (Normalized)

Destructive test for the evaluation of mechanical properties under **tensile load**

Material to test : SPECIMEN → uniform cross-section

Calibration length $\equiv$ Necessary length so that σ is transmitted uniformly

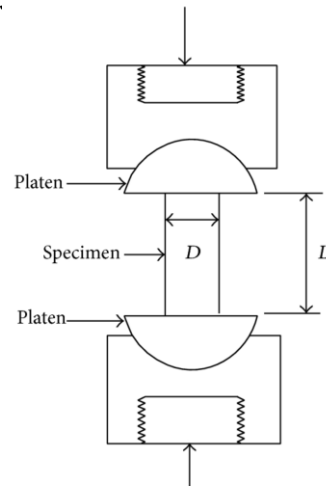


BASIC CONCEPTS

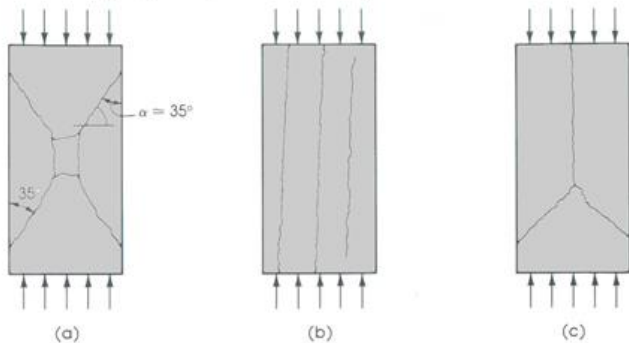
Compression test

Application to materials that exhibit a fragile behavior (i.e. concrete)

Widely applied to ceramics

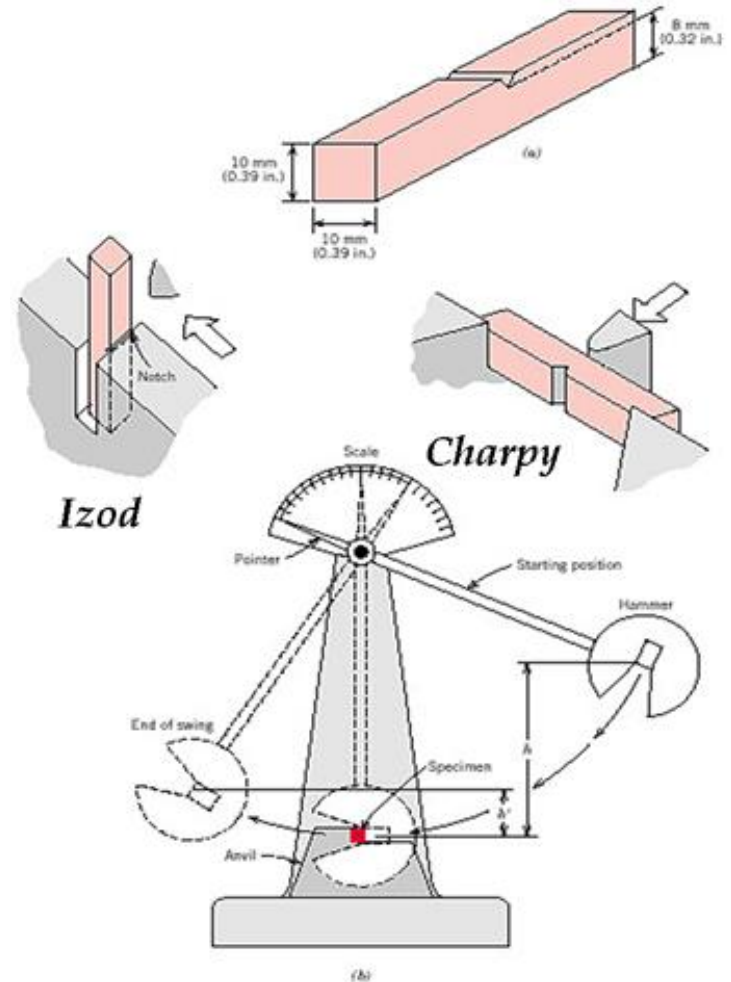


Typical failure modes in concrete



Impact test

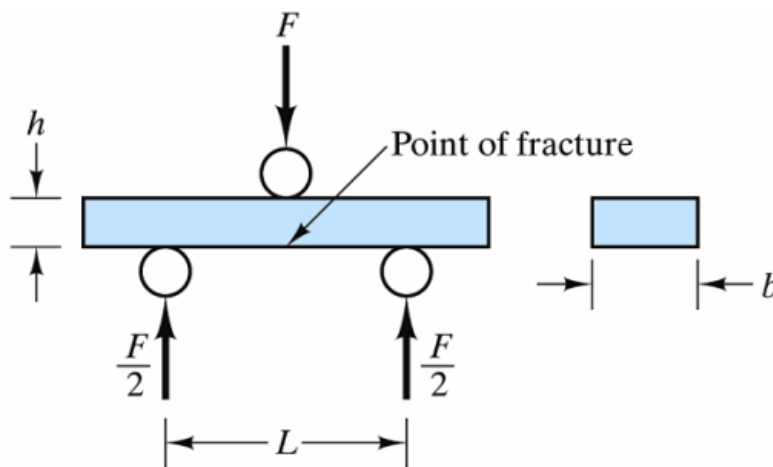
Determines toughness as the energy absorbed by a material until fracture



BASIC CONCEPTS

Three-point bend test

Determine the modulus of elasticity in bending, flexural stress, flexural strain and the flexural stress–strain response of the material.



At the point where load is applied:

- ✓ Upper surface is under compressive stress.
- ✓ Lower surface is under tensile stress.
- ✓ Fracture occurs along the outermost sample edge, which is under tensile stress

Modulus of Rupture:

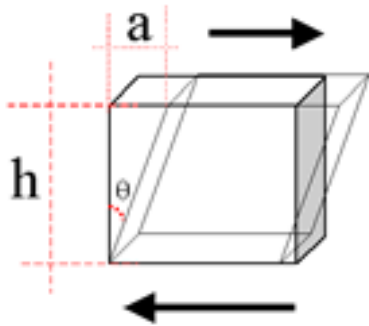
$$MOR = \frac{3 \cdot F \cdot L}{(2bh^2)}$$

Solves the dispersion in results obtained from tensile and compression tests because here the specimen is subjected to both: tension and compression

BASIC CONCEPTS

Shear test

A shear test is designed to apply stress to a test sample so that it experiences a sliding failure along a plane that is parallel to the forces applied.



Shear stress:

$$\tau = \frac{F}{A_0}$$

Shear strain:

$$\gamma = \frac{a}{h} = \tan \theta$$

In the elastic regime:

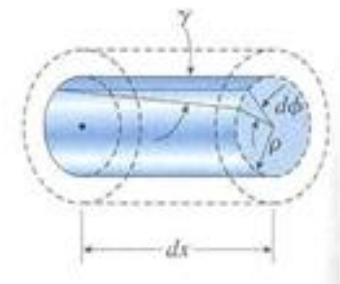
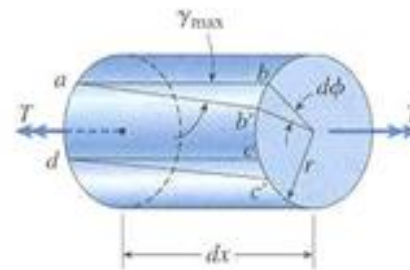
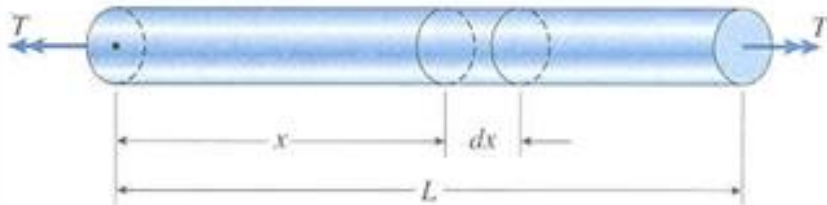
$$\tau = G \cdot \gamma$$

$$E = 2G(1 + \nu)$$

* G = Shear modulus

Torsion test

Is a variant of the shear test. A twisting force is applied onto a bar, creating a certain angle of rotation.



$$\tau = f(F_{\text{torsion}}) \quad \gamma = f(\phi)$$

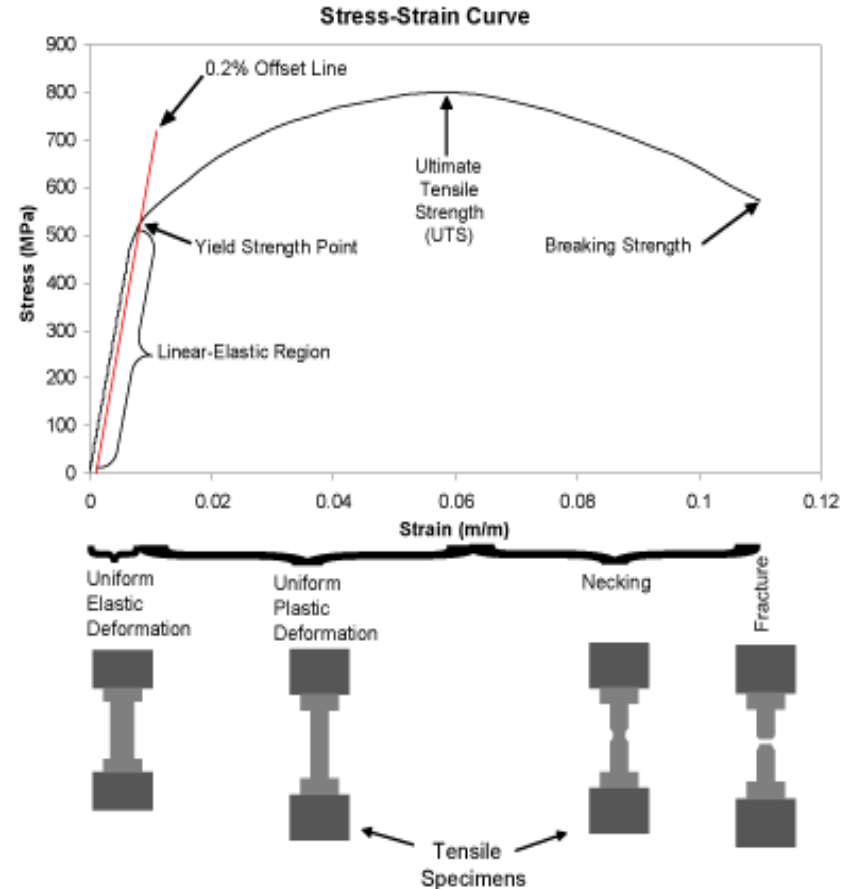
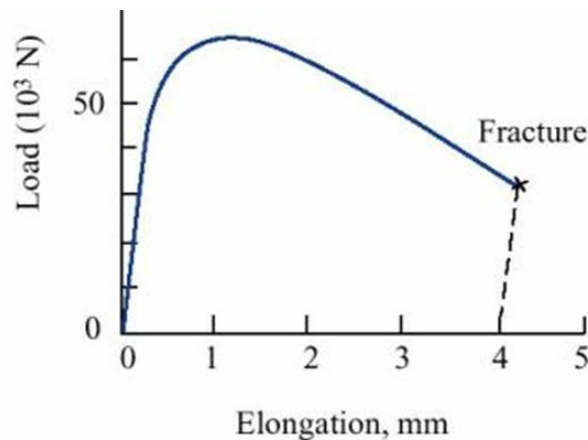
TOPIC 5. MECHANICAL PROPERTIES I

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UNIAXIAL TENSION TEST

What happens to the sample as we apply tensile force?

Output of test: **Stress-Strain curve**



Nominal (Engineering) stress (σ_n):

$$\sigma_n = \frac{F}{A_0}$$

MPa (10^6 N/m²)
psi (pounds/inch²)

Nominal (Engineering) strain (ϵ_n):

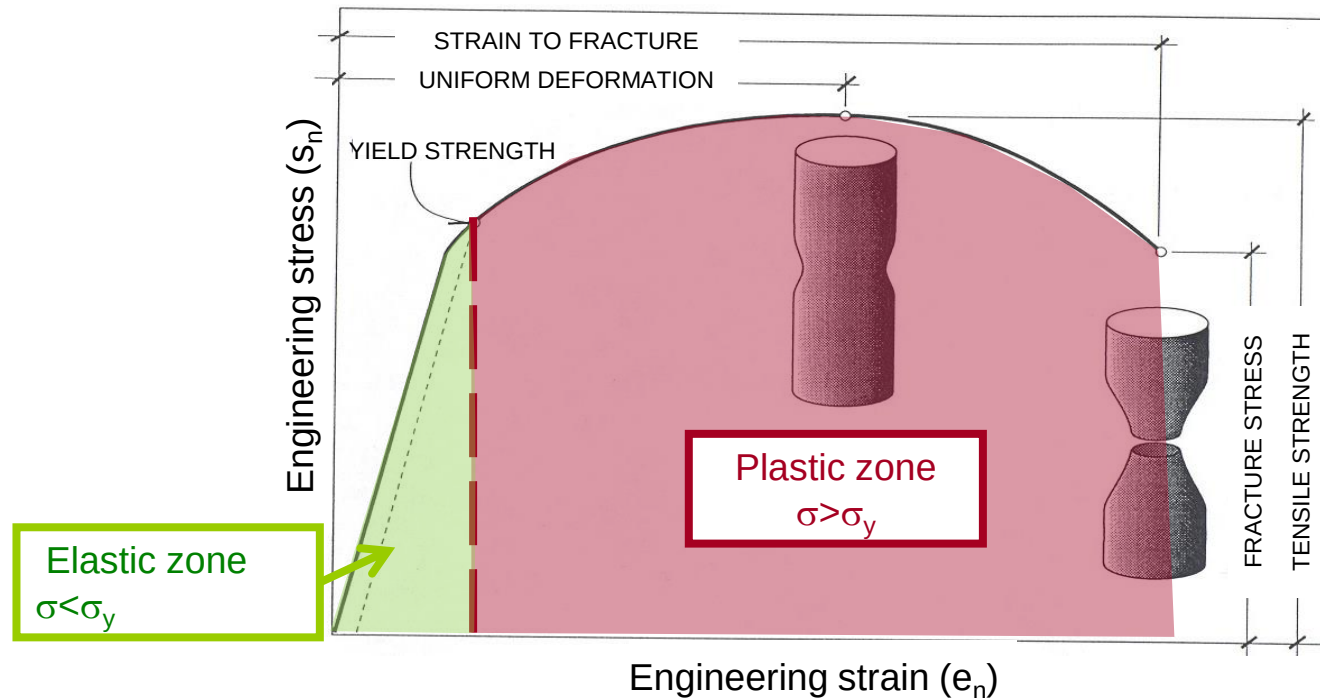
$$\epsilon_n = \frac{l - l_0}{l_0}$$

(non-dimensional)

F = Applied Load (Force)
 A_0 = Cross-sectional Area
 l = Length of specimen
 l_0 = Initial length of specimen

STRESS - STRAIN CURVE

Output of test: **Stress-Strain curve**



Elastic Deformation: upon unloading $\Rightarrow$ specimen recovers original form

Plastic Deformation: ϵ is no longer proportional to $\sigma \Rightarrow$ permanent deformation

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STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

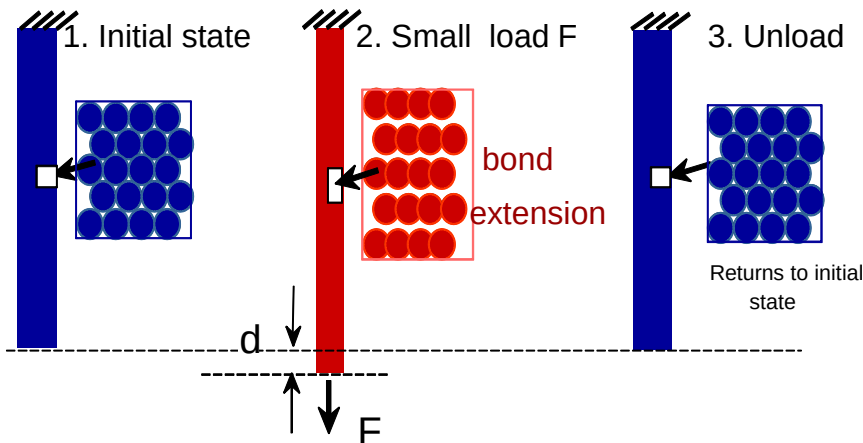
When the load is removed $\Rightarrow$ The specimen recovers its original shape

E: Young's Modulus or Elastic Modulus

Stiffness of material or resistance to elastic deformation

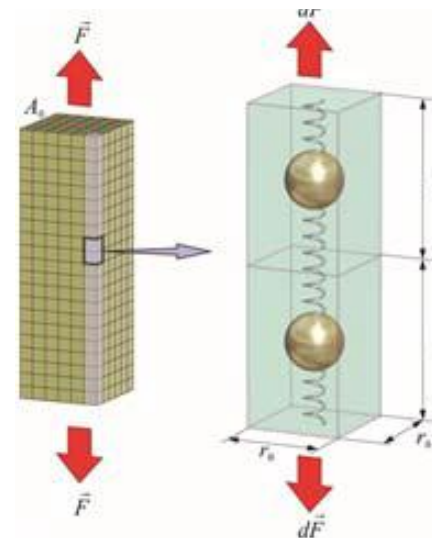
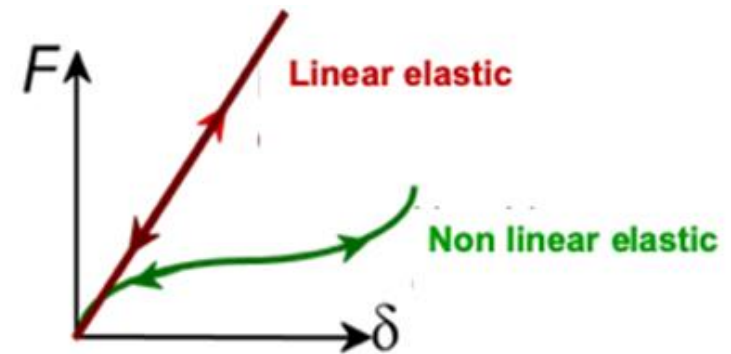
Elastic means reversible

Deformation mechanism



At atomic scale, macroscopic elastic deformation appears as small changes in the interatomic spacing and the bonds between atoms are stretched.

(atoms are displaced from their original position but they do not get to occupy others)



STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

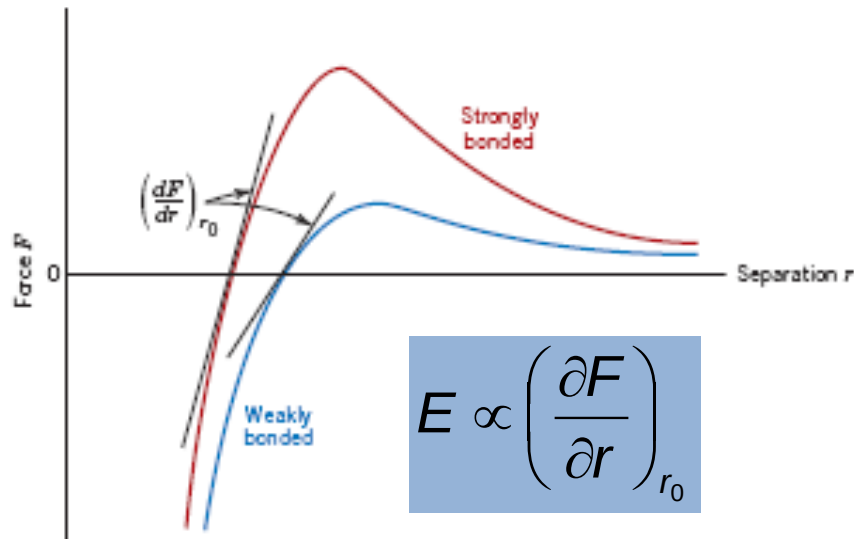
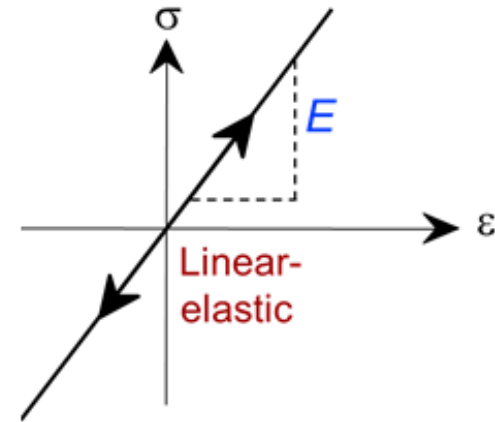
a) Linear

$$\sigma = E \varepsilon \quad \text{Hooke's law}$$

σ proportional to ε

$E = f(\text{bond strength}) \Rightarrow \text{if } \uparrow \text{ bond strength} \Rightarrow \uparrow E$

Mechanism associated to Elastic deformation = Bond relaxation.



$$E_{\text{ceramics}} > E_{\text{metals}} > E_{\text{polymers}}$$

Material	Modulus E (GPa)	Density ρ (Mg m^{-3})	Specific modulus E/ρ (GPa/Mg m^{-3})
Steels	210	7.8	27
Al alloys	70	2.7	26
Alumina, Al_2O_3	390	3.9	100
Silica, SiO_2	69	2.6	27
Cement	45	2.4	19

Bond force $\leftrightarrow$ T melting

$\uparrow$ Bond strength $\leftrightarrow \uparrow E$

STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

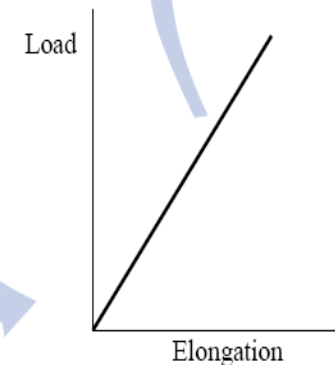
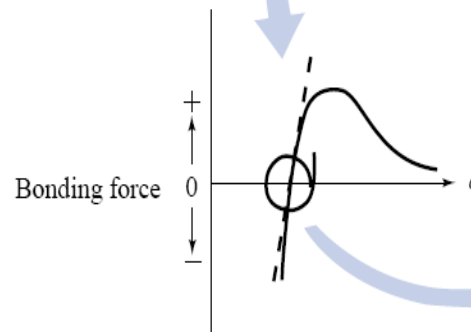
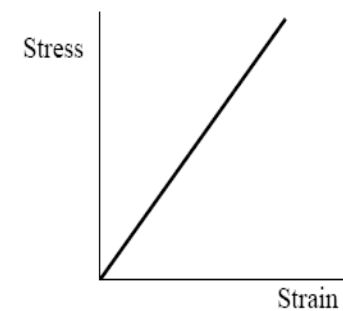
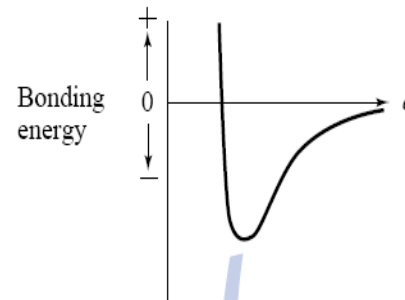
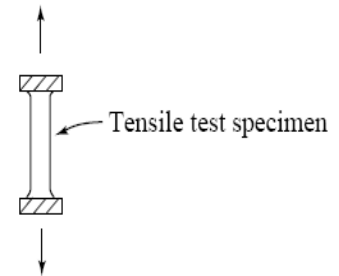
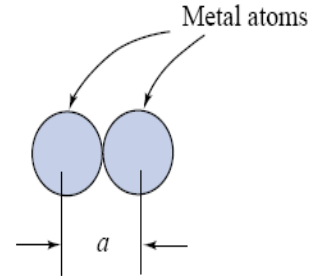
a) Linear

$$\sigma = E \varepsilon \quad \text{Hooke's law}$$

σ proportional to ε

The atoms tend to $\Rightarrow$ minimize energy.

Position of atoms = f (bond, attractive forces).



STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

b) Non linear

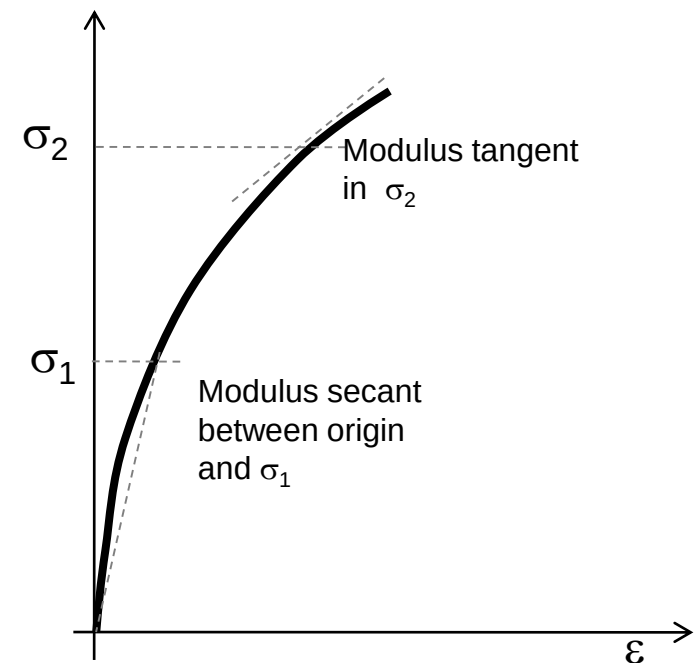
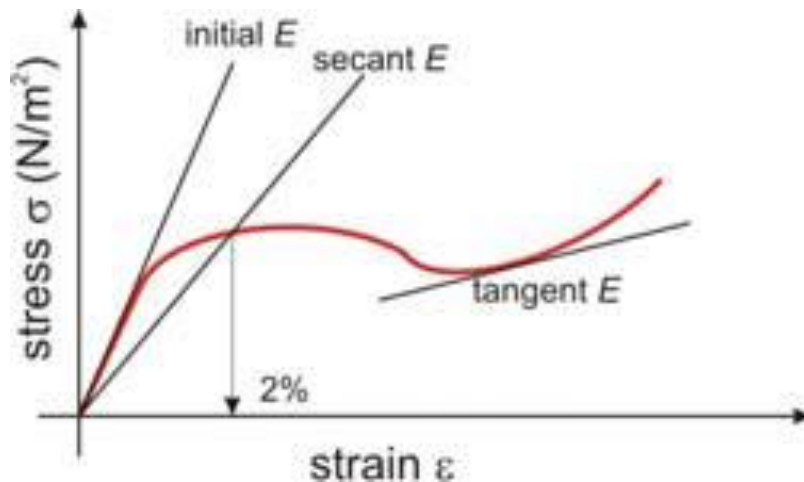
It is produced in some materials (some polymers, concrete, gray cast irons)

- E values depend on ε
- Hooke's law: $\sigma = E(\varepsilon) \varepsilon$
- Specific values of E are defined for practical purposes:

Initial E: $\varepsilon < 0.5\%$

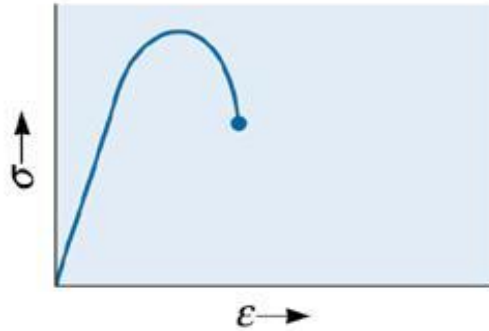
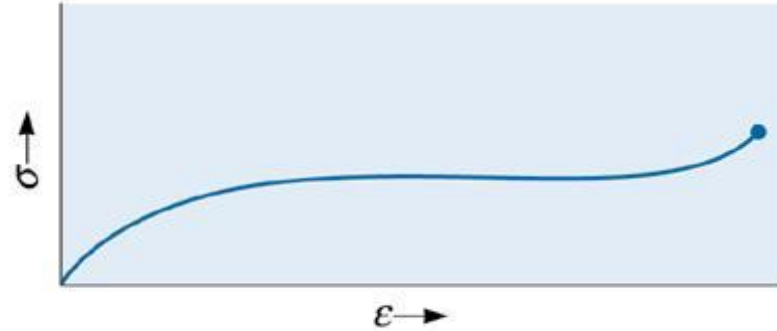
Secant E: $\varepsilon = 2\%$

Tangent E: at desired ε

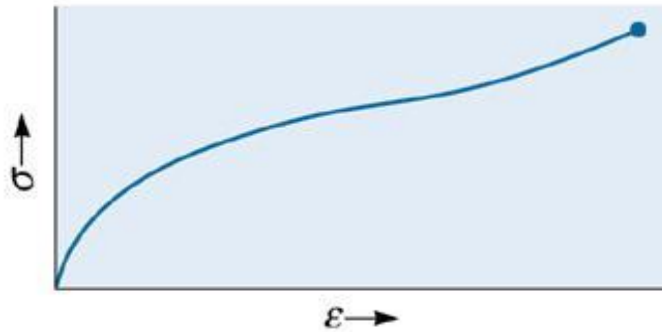


STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

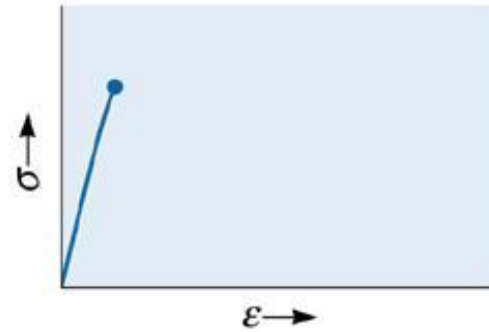
(a) Metal

(b) Thermoplastic material above T_g 

(c) Elastomer



(d) Ceramics, glasses, and concrete



Tensile stress-strain curves for different materials. Note that these are qualitative

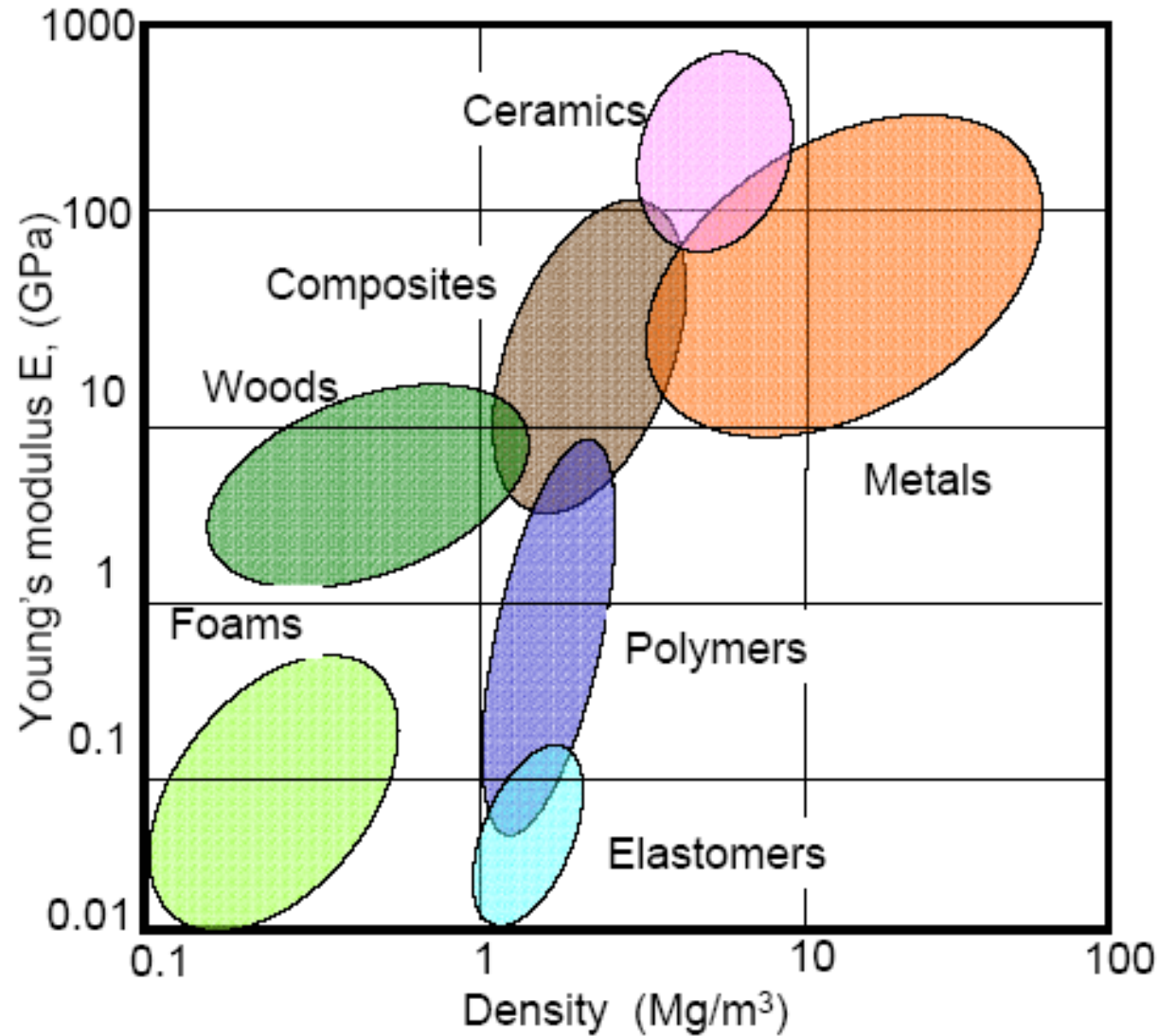
E VALUES FOR DIFFERENT MATERIALS

<i>Material</i>	<i>Modulus of Elasticity</i>	
	<i>GPa</i>	<i>10⁶ psi</i>
Metal Alloys		
Tungsten	407	59
Steel	207	30
Nickel	207	30
Titanium	107	15.5
Copper	110	16
Brass	97	14
Aluminum	69	10
Magnesium	45	6.5
Ceramic Materials		
Aluminum oxide (Al ₂ O ₃)	393	57
Silicon carbide (SiC)	345	50
Silicon nitride (Si ₃ N ₄)	304	44
Spinel (MgAl ₂ O ₄)	260	38
Magnesium oxide (MgO)	225	33
Zirconia ^a	205	30
Mullite (3Al ₂ O ₃ -2SiO ₂)	145	21
Glass-ceramic (Pyroceram)	120	17
Fused silica (SiO ₂)	73	11
Soda-lime glass	69	10

<i>Material</i>	<i>Modulus of Elasticity</i>	
	<i>GPa</i>	<i>10⁶ psi</i>
Polymers^b		
Phenol-formaldehyde	2.76–4.83	0.40–0.70
Polyvinyl chloride (PVC)	2.41–4.14	0.35–0.60
Polyester (PET)	2.76–4.14	0.40–0.60
Polystyrene (PS)	2.28–3.28	0.33–0.48
Polymethyl methacrylate (PMMA)	2.24–3.24	0.33–0.47
Polycarbonate (PC)	2.38	0.35
Nylon 6,6	1.58–3.80	0.23–0.55
Polypropylene (PP)	1.14–1.55	0.17–0.23
Polyethylene—high density (HDPE)	1.08	0.16
Polytetrafluoroethylene (PTFE)	0.40–0.55	0.058–0.080
Polyethylene—low density (LDPE)	0.17–0.28	0.025–0.041

E VALUES FOR DIFFERENT MATERIALS

Bubble chart



STRESS-STRAIN CURVE: ELASTIC ZONE $\sigma < \sigma_y$

Poisson's ratio $\nu \cong$ Relation between transverse and axial strain

All **elastic** longitudinal deformation $\Rightarrow$ Δ dimensional transverse

** Isotropic material*

If:

- Uniaxial stress applied
- Isotropic material

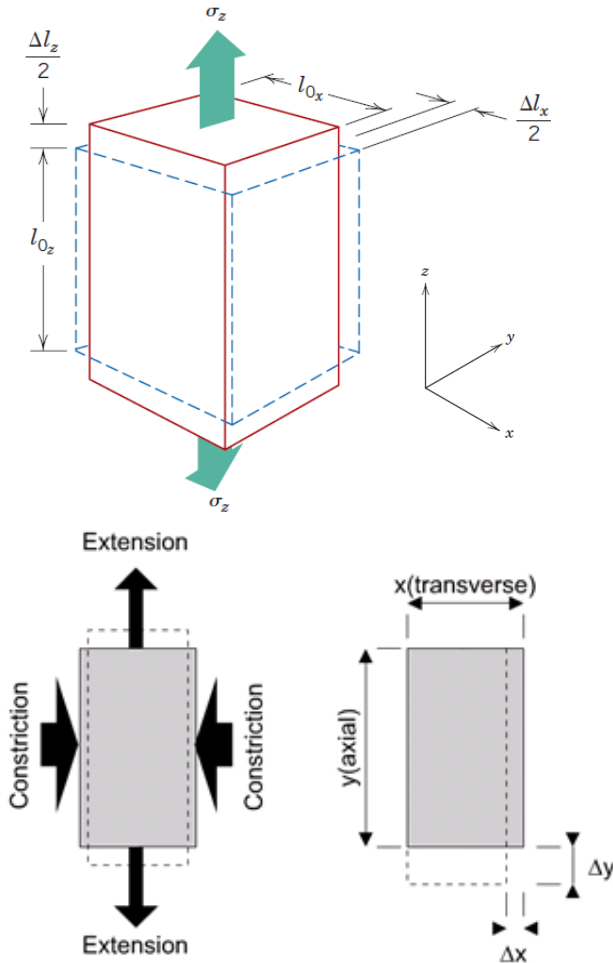
Then:

$$\nu = \frac{-\varepsilon_{\text{transverse}}}{\varepsilon_{\text{longitudinal}}} = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$

Ideal materials: $\nu=0.5$ (perfect incompressible materials $\rightarrow$ no change in volume)

$\nu = -1 \rightarrow$ no change in aspect ratio

Real materials : $\nu=0.2-0.4$



	ν
natural rubber	0,499-0,49
Polymers	0,3-0,45
Metals	0,25-0,4

TOPIC 5. MECHANICAL PROPERTIES I

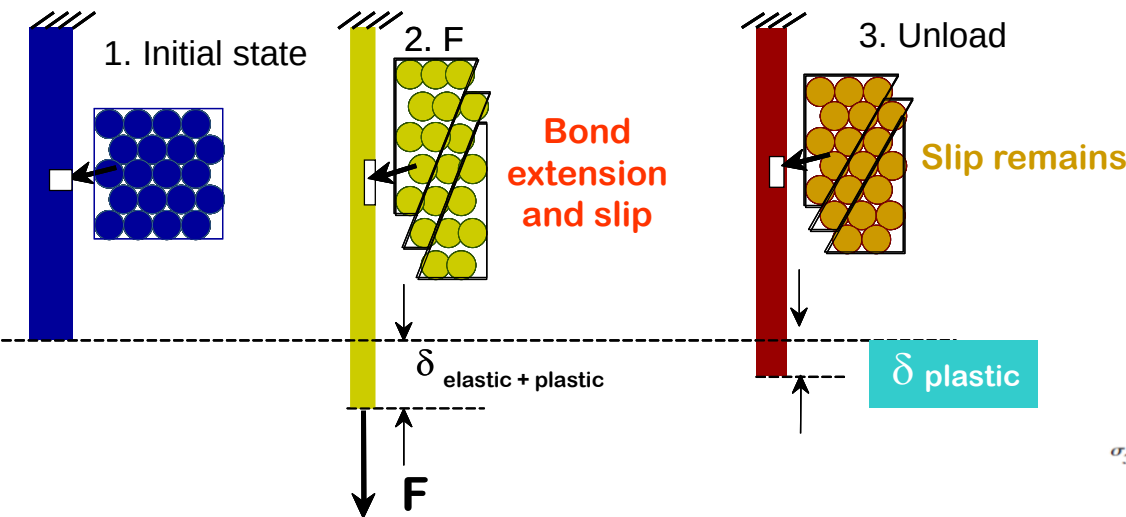
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STRESS-STRAIN CURVE: PLASTIC ZONE $\sigma > \sigma_y$

ϵ is **not** proportional to $\sigma \Rightarrow$ Permanent deformation

• If $F=0 \Rightarrow \epsilon > 0$

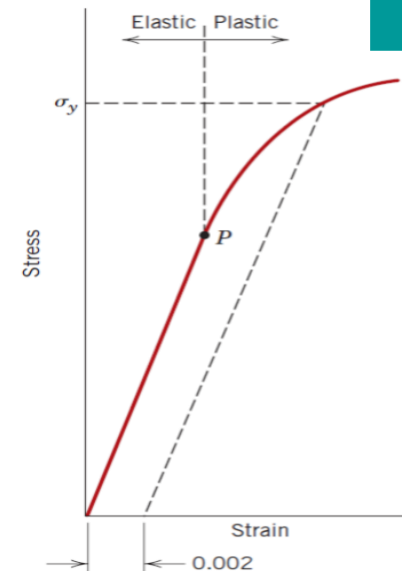
Breaking and formation of new bonds $\Rightarrow$ no recovering of initial form



Plastic means permanent !

The atoms are displaced permanently from their initial positions to new positions

In the majority of metals $\epsilon_{\text{plastic max.}} = 0,5\%$



Dislocations movement

Forming through deformation

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PLASTIC DEFORMATION: SLIPPING OF DISLOCATIONS

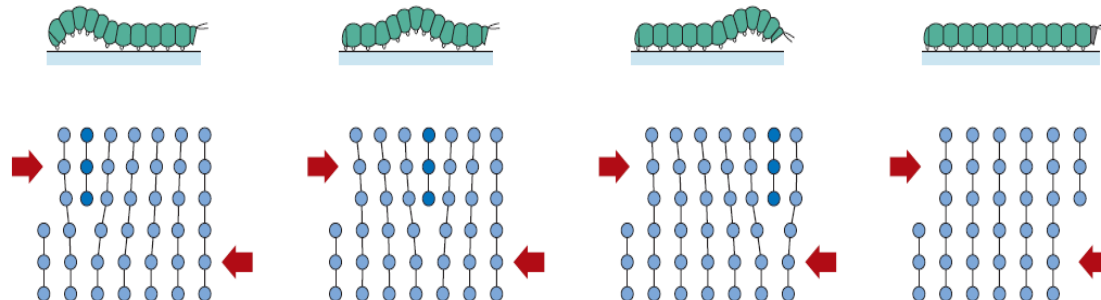
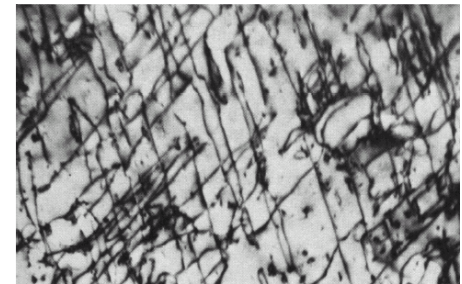
Slipping of dislocations

- ϵ_p without dislocations: we have to break simultaneously various bonds and remake them through slipping.
- ϵ_p with dislocations: The breakage and formation of bonds is sequential → Lower energy consumption → When dislocations are moving, it is easier to glide atoms

Consequences: they could be displaced inside the crystal with relatively low forces and produce the complete displacement on crystalline planes .

It explains:

- Plastic deformation in metals (workability and ductility)
- $E_{\text{theoretical}} > E_{\text{exp}}$.



Slip system:
Slip plane + slip direction

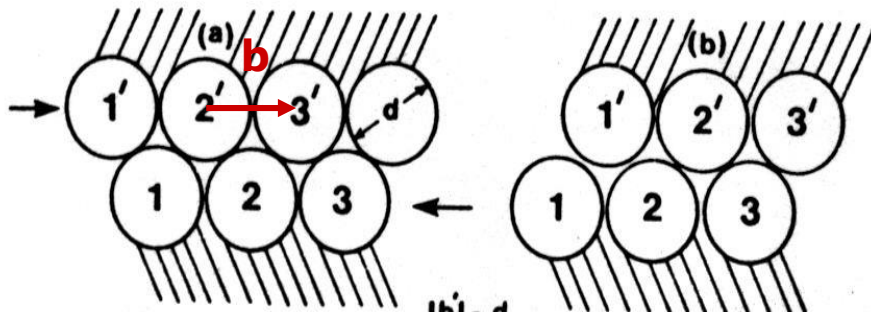
PLASTIC DEFORMATION: SLIPPING OF DISLOCATIONS

Dislocation movement **is parallel to the direction of maximum packing** $\Rightarrow$ higher distance between planes $\Rightarrow$ lower F_{atr} between them $\Rightarrow$ easier slipping

$$E \text{ used to move a dislocation} = E \propto |b|^2$$

Compact direction

FCC $_{[110]}$

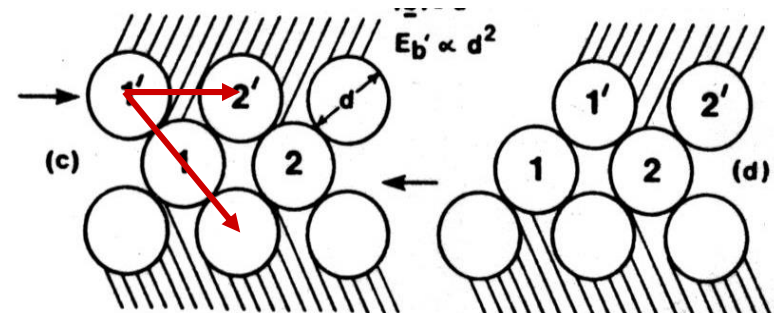


$$b = 2 \cdot R$$

$$E_b \propto 4R^2$$

Non compact direction

FCC $_{[100]}$



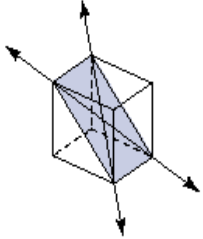
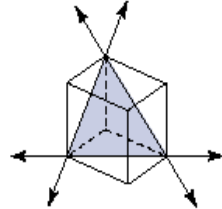
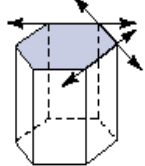
$$b^2 + b^2 = (4R)^2 \Rightarrow b = 2\sqrt{2} \cdot R$$

$$E_b \propto 8R^2$$

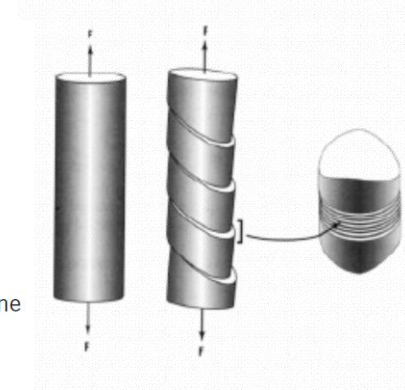
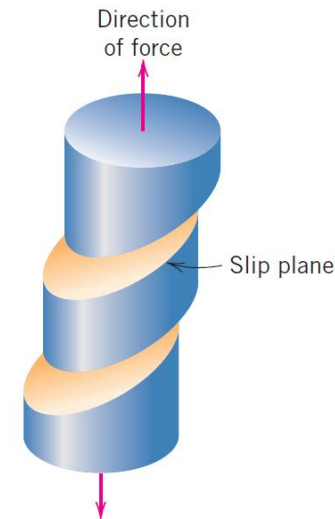
PLASTIC DEFORMATION: SLIP SYSTEMS

- Permanent plastic deformation takes place when the atoms slip between them.
- Slip occurs along specific crystallographic planes that depend on the structure

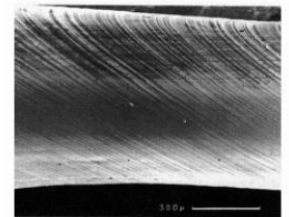
Slipping directions and planes $\Rightarrow$ SLIPPING SYSTEMS

Crystal structure	Slip plane	Slip direction	Number of slip systems	Unit cell geometry	Examples
bcc	{110}	$\langle \bar{1}11 \rangle$	$6 \times 2 = 12$		α -Fe, Mo, W
fcc	{111}	$\langle \bar{1}10 \rangle$	$4 \times 3 = 12$		Al, Cu, γ -Fe, Ni
hcp	(0001)	$\langle 110 \rangle$	$2 \times 3 = 6$		Cd, Mg, α -Ti, Zn

Example: Deformation of a Cd single crystal (hcp)



Slip bands



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STRESS - STRAIN CURVE

Nominal (Engineering) stress (σ_n):

$$\sigma_n = \frac{F}{A_0}$$

MPa (10^6N/m^2)

psi (pounds/inch²)

Nominal (Engineering) strain (ϵ_n):

$$\epsilon_n = \frac{l - l_0}{l_0}$$

(non-dimensional)

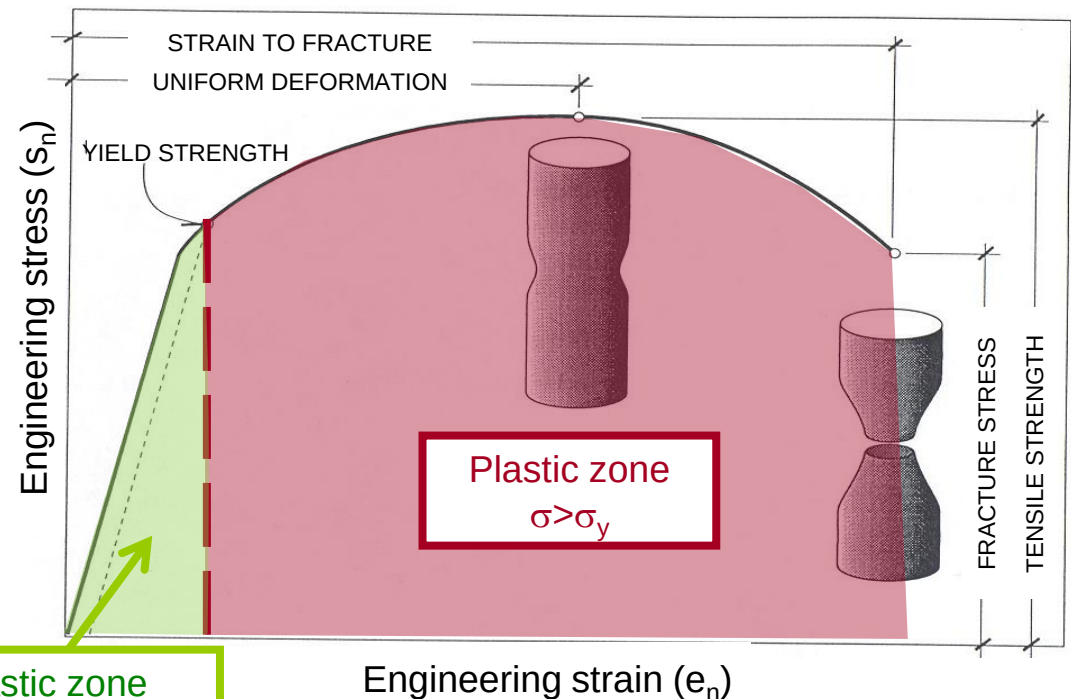
Modulus of elasticity:

$$E = \frac{\sigma}{\epsilon}$$

CURVE PARAMETERS

Young's modulus
Yield stress (σ_y)
Strain
Tensile Strength
Toughness
Resilience

F = Applied Load (Force)
 A_0 = Cross-sectional Area
 l = Length of specimen
 l_0 = Initial length of specimen



Elastic zone
 $\sigma < \sigma_y$

STRESS-STRAIN CURVE: CALCULATION OF THE YIELD STRESS

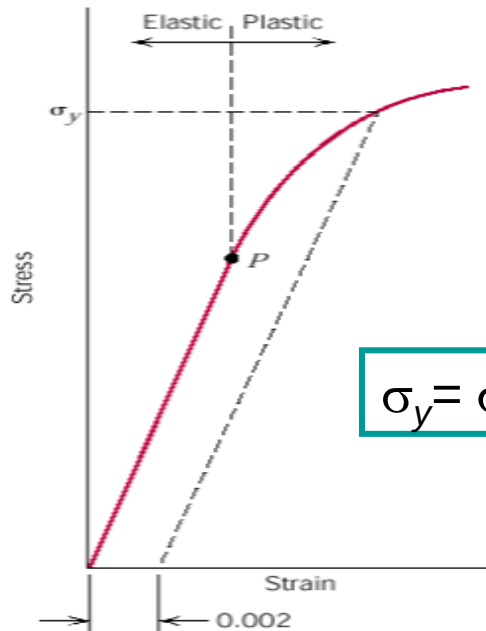
Yield strength (stress) σ_y : stress above which **permanent** deformation occur

Criteria for the calculation of the Yield Stress

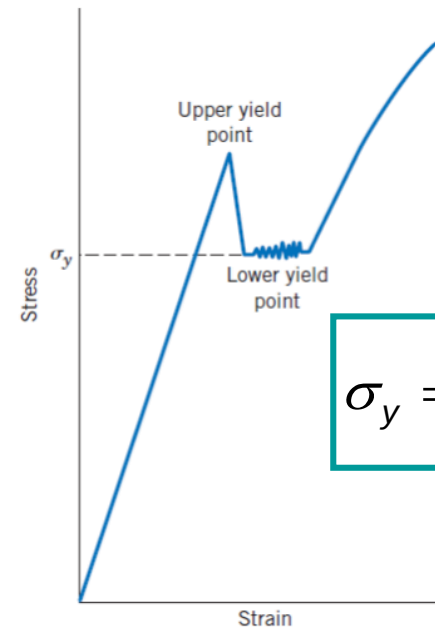
Depending on the deformation of material we adopt $\neq$ criteria

Conventional Yield Stress: The most frequently used in metals $\approx 0,2\%$ of ϵ

Discontinuity phenomena in the yield point:
Low carbon steels ,some polymers , etc...



$$\sigma_y = \sigma (0,2\% \text{ of } \epsilon)$$



$$\sigma_y = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

STRESS-STRAIN CURVE: SECURITY FACTOR

Due to the variability in mechanical properties

Design $\Rightarrow \sigma_{\text{security}} (\sigma_w)$ working stress

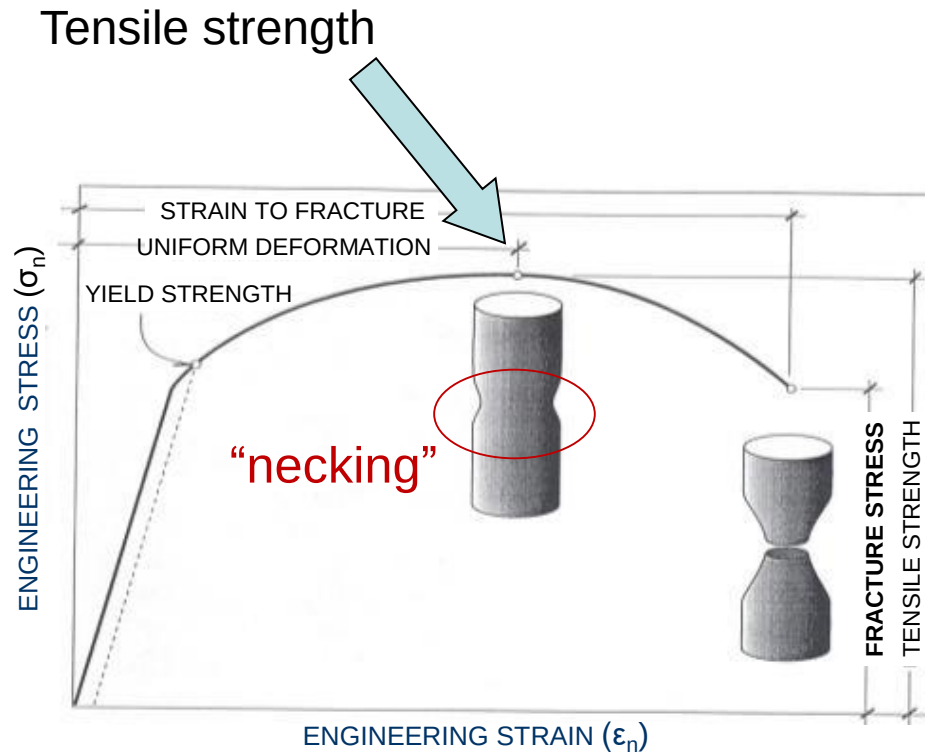
$$\sigma_w = \frac{\sigma_y}{N}$$

σ_y = yield strength

$N = 1.2 - 4$ (+ usually 2)

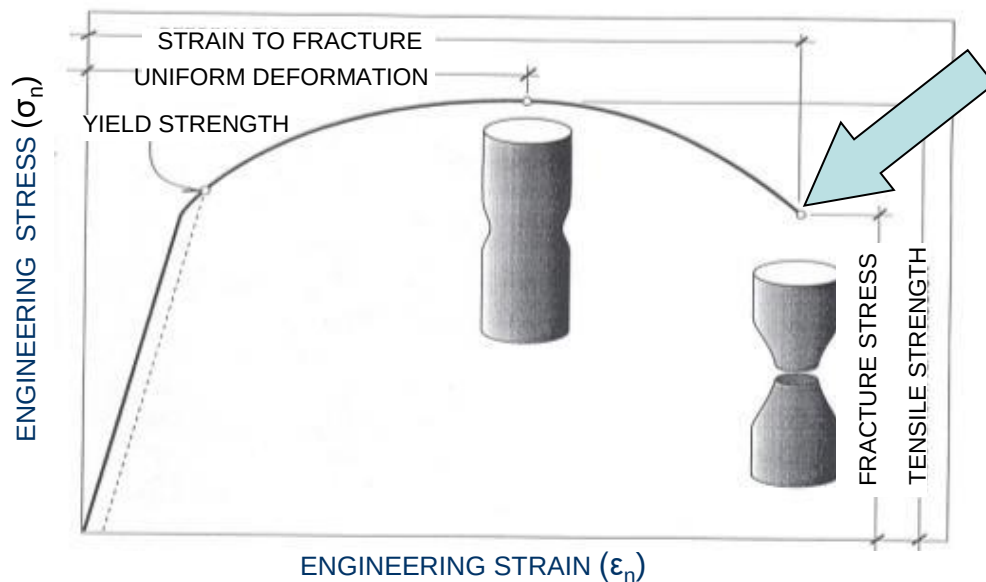
STRESS-STRAIN CURVE: CALCULATION OF UTS AND σ_{fracture}

Tensile strength (or ultimate TS): maximum nominal stress that a specimen supports under tension

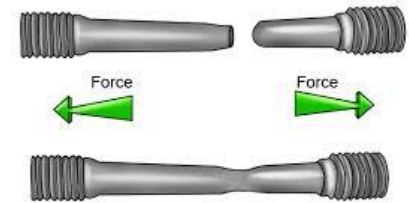


STRESS-STRAIN CURVE: CALCULATION OF UTS AND σ_{fracture}

Fracture stress: stress value at fracture



Fracture stress



The more ductile the material, the higher the amount of “necking” or constriction,

The σ_{fracture} gives an idea of the presence of defects (porosity, inclusions) in the material if the value is very low

STRESS-STRAIN CURVE: DUCTILITY

Ductility: Measure of the degree of plastic deformation grade that can be supported by a material before fracture, $f(T)$

% Elongation after failure:

Percentage of plastic strain at fracture (removing elastic recovery)

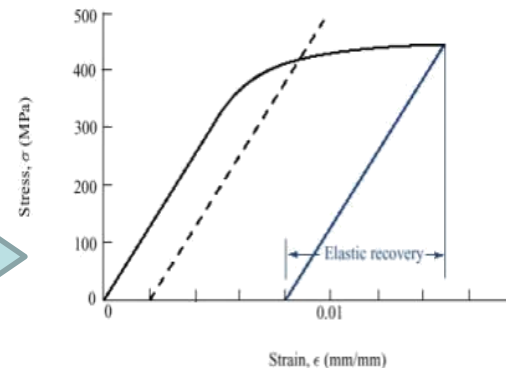
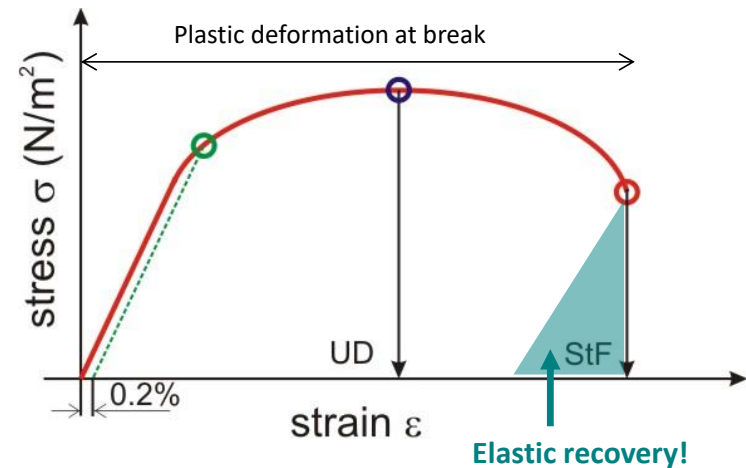
$$EL(\%) = \frac{l_f - l_0}{l_0} \times 100$$

l_0 is the initial gauge length between marks
 l_f is the final length between marks after fracture

It is measured once the specimen has fractured → The material has recovered the elastic deformation

Before fracture $F \neq 0$, $\epsilon_t = \epsilon_e + \epsilon_p$

After fracture $F=0$, $\epsilon_t = \epsilon_p$



UD: Uniform deformation ()
StF: Strain to fracture (StF)
Plastic strain $\epsilon_p=0.2\%$: for determination of yield strength

STRESS-STRAIN CURVE: DUCTILITY

Percent reduction in area (constriction)

$$RA(\%) = \frac{A_0 - A_f}{A_0} \times 100$$

It is measured when the specimen has fractured.

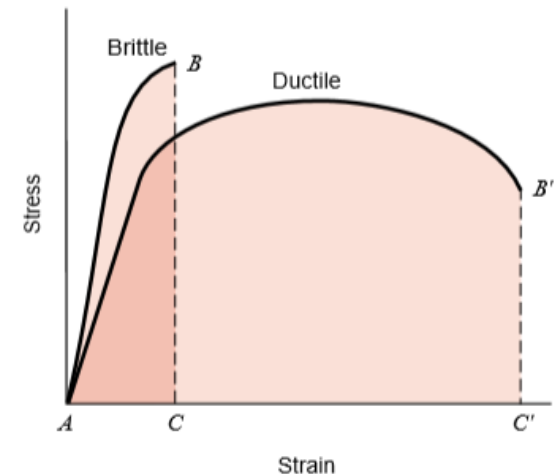
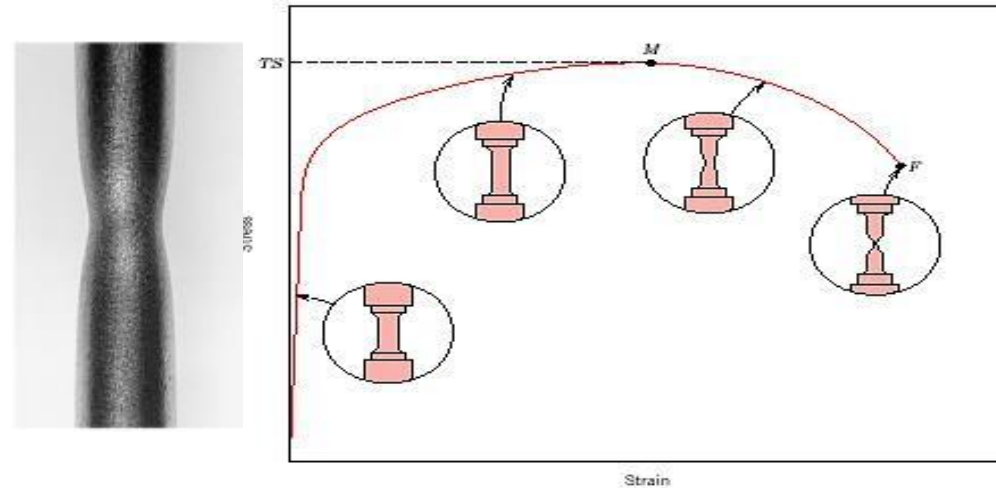
A_0 initial cross sectional area.

A_f final cross sectional area (measured after fracture)

Normally $\varepsilon (\%) \neq AR(\%)$

Ductility can be measured as a percentage of elongation or as percentage in reduction of area.

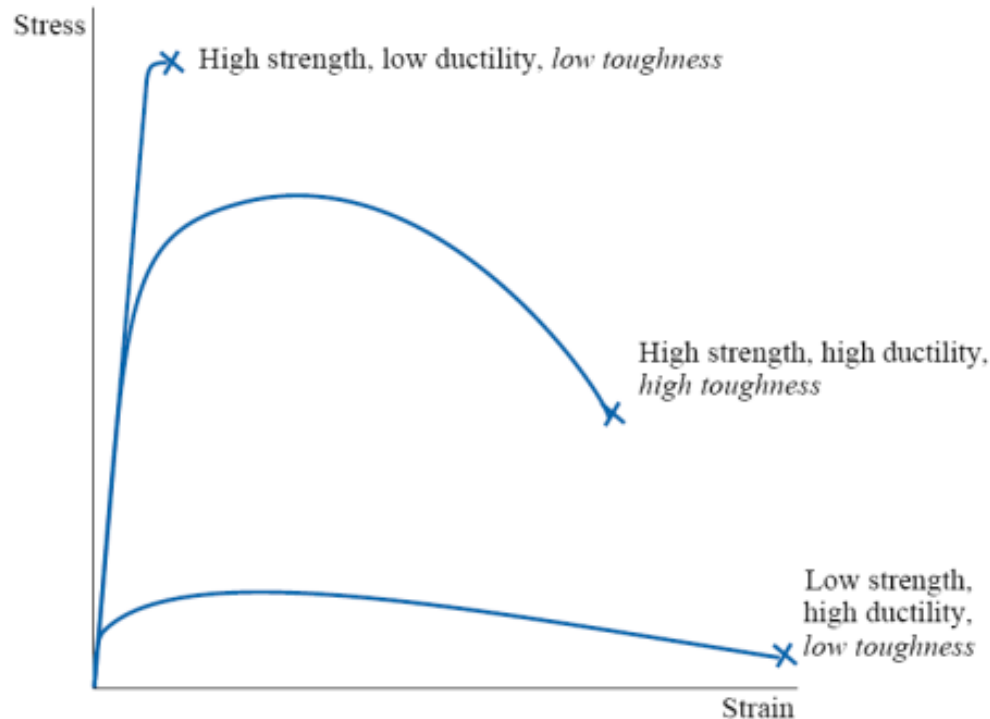
It indicates the maximum deformation the material can withstand before fracture → how many forming operations can be sequentially performed (i.e. lamination)



STRESS-STRAIN CURVE: TOUGHNESS

Toughness: Ability of a material to absorb energy up to fracture, by elastic and plastic deformation

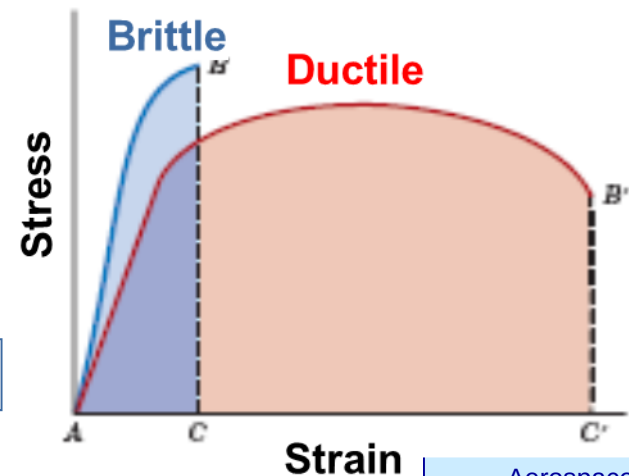
$E_{\text{elastic+plastic deformation}}$ per unit Volume (Pa or J/m³)



- Important in structural applications
- Ability to absorb energy
(protection and security systems, automotion)



- Fracture type (ductile, brittle)
- Fracture toughness

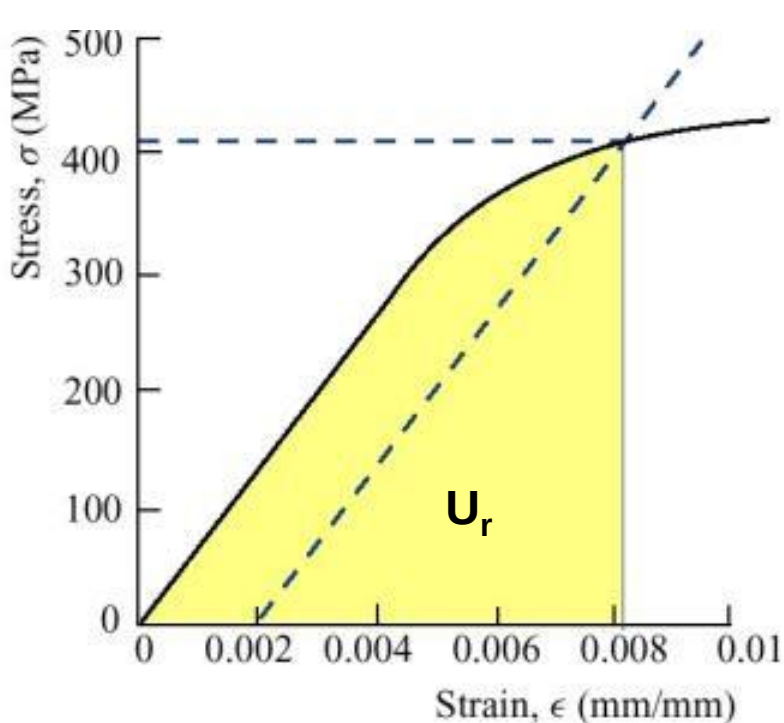


Area under the curve: It is a measure of **toughness**

STRESS-STRAIN CURVE : RESILIENCE

Resilience: Capacity of a material to absorb E_{elastic} when it is deformed **elastically**

Modulus of Resilience U_r : $E_{\text{deformation}}$ per unit Volume (Pa or J/m³) necessary to deform a material up to the yield point



Application: alloys to fabricate springs

Area under the curve $U_r = \int_0^{\epsilon_y} \sigma d\epsilon$

Integrating, if the elastic region is linear

$$U_r = \frac{1}{2} \sigma_y \epsilon_y$$

Substituting $\epsilon_y = \frac{\sigma_y}{E}$ (Hooke's law)

$$U_r = \frac{1}{2} \frac{\sigma_y^2}{E}$$

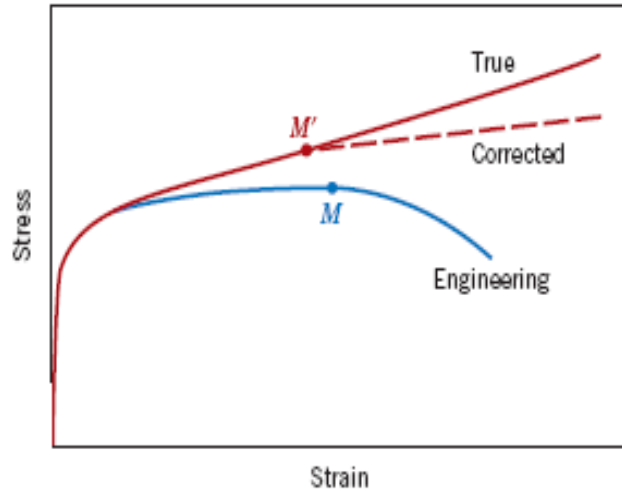
Materials with $\uparrow$ Resilience:
 $\uparrow \sigma_y \downarrow E$

TOPIC 5. MECHANICAL PROPERTIES I

- Basic Concepts
- Uniaxial Tension Test
- Stress-strain Curve: Elastic Zone
- Stress-strain Curve: Plastic Zone
- Slip Systems
- Stress-strain Curve: Calculations
- **True Stress-strain Curves**

TRUE STRESS-STRAIN CURVES

Original dimensions change during the test $\Rightarrow \sigma_{\text{real}} > \sigma_{\text{eng}}$ and $\epsilon_{\text{real}} < \epsilon_{\text{eng}}$



Engineering stress (σ_r): $\sigma_{\text{eng}} = \frac{F}{A_0}$

True or real stress (σ_r): $\sigma_r = \frac{F}{A_i} = \frac{F}{A_0} \frac{A_0}{A_i} = \sigma_{\text{eng}} \frac{A_0}{A_i}$

True Strain: when the strains are large (plastic zone) the longitudinal base changes during the test.

$$\epsilon_r = \frac{L_1 - L_0}{L_0} + \frac{L_2 - L_1}{L_1} + \frac{L_3 - L_2}{L_2} + \dots \Rightarrow \epsilon_r = \int_{L_0}^L \frac{dL}{L} = \ln \frac{L}{L_0}$$

TRUE STRESS-STRAIN CURVES

Before necking (uniform deformation)

The volume affected remains the same $\Delta V = 0 \Rightarrow$ it is true that:

$$l_0 A_0 = l_i A_i \longrightarrow \frac{l_i}{l_0} = \frac{A_0}{A_i}$$

$$\sigma_r = \sigma_{eng} \frac{A_0}{A_i} = \sigma_{eng} \frac{l_i}{l_0}$$

Since $l = l_0 + \Delta l$:

$$\epsilon_r = \ln(1 + \epsilon_{eng})$$

$$\sigma_r = \sigma_{eng} (1 + \epsilon_{eng})$$

(until the beginning of necking)

After necking (deformation is not uniform)

$$\sigma = \frac{F_i}{A_i} = \sigma_{eng} \frac{A_0}{A_i}$$

$$\epsilon_r = \ln \frac{A_0}{A} = 2 \ln \frac{D_0}{D}$$

A_i : instantaneous cross-sectional area

A_0 : initial cross-sectional area

D_0 : initial diameter

D : instantaneous diameter

SUMMARY

https://www.youtube.com/watch?v=BHZALtqAjeM&ab_channel=RealEngineering

